Objectives:

1. To optimize efficiency by minimizing energy losses and ensuring effective power transfer. To regulate voltage and maintain stable output under varying load conditions.
2. To design a compact and lightweight transformer that reduces material usage and saves space.
3. To reduce manufacturing and operational costs through the efficient use of shared windings and fewer components.
4. To ensure the transformer handles specific load requirements without saturation or overloading.
5. To provide customizable voltage taps for flexibility in input and output voltage levels.
6. To use eco-friendly materials and sustainable practices to reduce environmental impact.

Introduction:

An auto transformer or autotransformer is a type of electrical transformer that has a single winding that serves both as a primary winding as well as a secondary winding.

In simple terms, an autotransformer is a transformer that has only one winding with taps brought out at different points along the winding.

The concept of an autotransformer is unique in the sense that the primary and secondary windings are directly connected to each other internally, unlike a regular isolation transformer where the windings are kept completely isolated.

The auto transformer therefore provides a special type of voltage transformation that is different from a standard isolation transformer.

Operation

An autotransformer has a single winding with two end terminals and one or more terminals at intermediate tap points. It is a transformer in which the primary and secondary coils have part of their turns in common. The portion of the winding shared by both the primary and secondary is the common section. The portion of the winding not shared by both the primary and secondary is the series section. The primary voltage is applied across two of the terminals. The secondary voltage is taken from two terminals, one terminal of which is usually in common with a primary voltage terminal.[3] Since the volts-per-turn is the same in both windings, each develops a voltage in proportion to its number of turns. In an autotransformer, part of the output current flows directly from the input to the output (through the series section), and only part is transferred inductively (through the common section), allowing a smaller, lighter, cheaper core to be used as well as requiring only a single winding. However, the voltage and current ratio of autotransformers can be formulated the same as other two-winding transformers:

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = a$$

Where $0 < V_2 < V_1$.

The ampere-turns provided by the series section of the winding:

$$F_S = (N_1 - N_2)I_1 = \left(1 - \frac{1}{a}\right) N_1 I_1$$

The ampere-turns provided by the common section of the winding:

$$F_C = N_2(I_2 - I_1) = \frac{N_1}{a}(I_2 - I_1)$$

For ampere-turn balance, $F_S = F_C$:

.

$$\left(1 - \frac{1}{a}\right) N_1 I_1 = \frac{N_1}{a}(I_2 - I_1)$$

Therefore:

$$\frac{I_1}{I_2} = \frac{1}{a} = \frac{N_2}{N_1}$$

One end of the winding is usually connected in common to both the voltage source and the load. The other end of the source and load are connected to taps along the winding. Different taps on the winding correspond to different voltages, measured from the common end. In a step-down transformer the source is usually connected across the entire winding while the load is connected by a tap across only a portion of the winding. In a step-up transformer, conversely, the load is attached across the full winding while the source is connected to a tap across a portion of the winding. For a step-up transformer, the subscripts in the above equations are reversed where, in this situation, N_2 and V_2 are greater than N_1 and V_1 respectively.

As in a two-winding transformer, the ratio of secondary to primary voltages is equal to the ratio of the number of turns of the winding they connect to. For example, connecting the load between the middle of the winding and the common terminal end of the winding of the autotransformer will result in the output load voltage being 50% of the primary voltage. Depending on the application, that portion of the winding used solely in the higher-voltage (lower current) portion may be wound with wire of a smaller gauge, though the entire winding is directly connected.

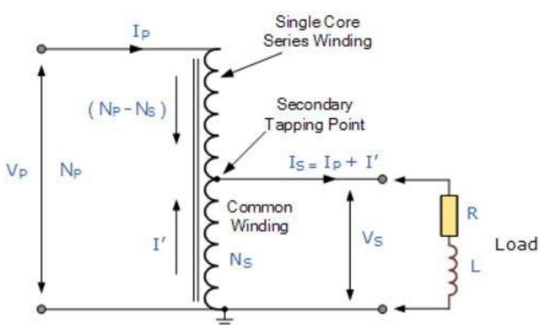
An autotransformer does not provide electrical isolation between its windings as an ordinary transformer does; if the neutral side of the input is not at ground voltage, the neutral side of the output will not be either. A failure of the isolation of the windings of an autotransformer can result in full input voltage applied to the output. Also, a break in the part of the winding that is used as both primary and secondary will result in the transformer acting as an inductor in series with the load (which under light load conditions may result in nearly full input voltage being applied to the output). These are important safety considerations when deciding to use an autotransformer in a given application.

Because it requires both fewer windings and a smaller core, an autotransformer for power applications is typically lighter and less costly than a two-winding transformer, up to a voltage ratio of about 3:1; beyond that range, a two-winding transformer is usually more economical.

In three-phase power transmission applications, autotransformers have the limitations of not suppressing harmonic currents and acting as another source of ground fault currents. A large three-phase autotransformer may have a "buried" delta winding, not connected to the outside of the tank, to absorb some harmonic currents.

In practice, losses mean that both standard transformers and autotransformers are not perfectly reversible; one designed for stepping down a voltage will deliver slightly less voltage than required if it is used to step up. The difference is usually slight enough to allow reversal where the actual voltage level is not critical.

Like multiple-winding transformers, autotransformers use time-varying magnetic field to transfer power. They require alternating currents to operate properly and will not function on direct current. Because the primary and secondary windings are electrically connected, an autotransformer will allow current to flow between windings and therefore does not provide AC or DC isolation.



$V_p = V_1$, $V_s = V_2$. $I_p = I_1$, $I_s = I_2$

Figure 1: Step Down Autotransformer Circuit Diagram

Our Design specifications:

As a part of our lab project we have designed an auto transformer. Our transformer's design specifications are given below

- Single-phase shell autotransformer
- Total apparent power: 80VA
- Intermittent operation
- Voltages: 210 /22V
- Frequency: 50Hz
- Cooling: air

COMPONETS USED IN THE DESIGN:

CORE:

Model: E-1114H

Material: Silicon steel (silicon content 3.5%)

$B_{MAX} = 1.2$ T

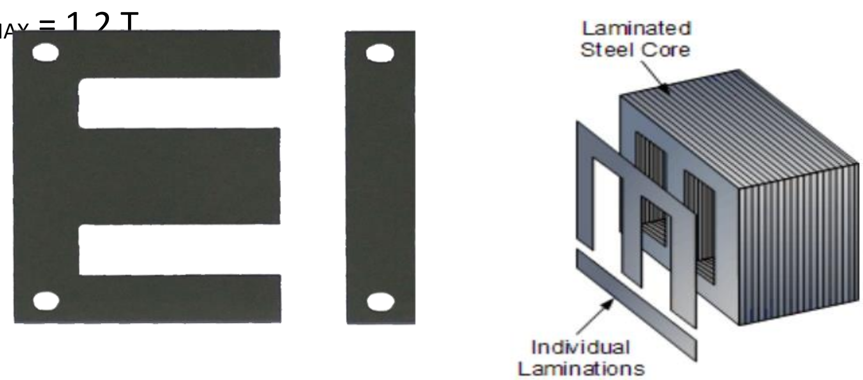


Figure 2: Transformer Core Lamination

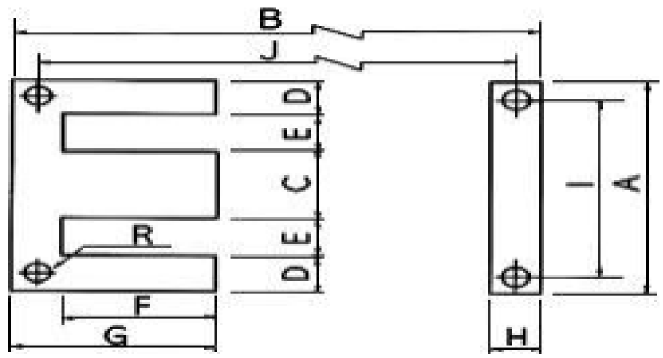


Fig 3: core lamination dimensions

Here,

A=111mm

B= 96.5mm

C= 35mm

D= 17.5 mm

E= 20.5mm

F = 55.5 mm

G = 76 mm

H = 20.5 mm

J = 76 mm

K = 10.25 mm

N = 95 mm

R = 7 mm

Properties:

- High permeability: Facilitates efficient magnetic flux flow.
- Low hysteresis loss: Reduces energy losses during magnetization and demagnetization cycles.
- High electrical resistivity: Minimizes eddy current losses.

Reel:

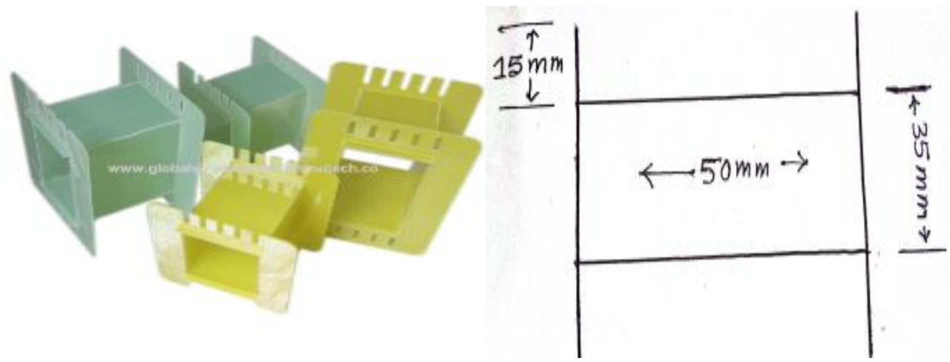


Fig 4 : Transformer reel

Here,

Internal space depth = 32.4 mm

Width = 35 mm

Length = 50 mm

Reel space available for winding layer = 15 mm

Calculation:

Design a 80 VA, 210V/24 V, 50 Hz, single phase auto-transformer. Do not consider the voltage drop at secondary side and assume the transformer efficiency of 90%.

Solution:

A_{ir} =Core cross-sectional area$= column\ width \times core\ depth$

$$= 35 \times 32.357\ mm^2$$

$$= 1132.5\ mm^2$$

For **E-1114H** Core (Silicon Steel) B_{max}=1.2 T

$$Air = \frac{\emptyset}{B} => \emptyset = Air \times B = 1132.5 \times 1.2 \times 10^{-6} = 1.359 \times 10^{-3} wb$$

$$\emptyset = Ku \times \sqrt{\frac{KVA}{f}}$$

$$=> 1.359 \times 10^{-3} = 3.6 \times 10^{-2} \times \sqrt{\frac{KVA}{50}}$$

$$=> KVA(A_d) = 71.2727\, VA$$

Conductive power, $A_d = A \times \frac{\text{Vmax}-\text{Vmin}}{\text{Vmax}} = A \times \frac{210-22}{210}$

$$=> 71.2727 = A \times \frac{210-22}{210}$$

$$=> A = 80\, VA$$

No. of Lamination= $\frac{32.357}{0.5} \cong 64$

Induced Voltage per Turn=4.44$\times f \times \emptyset$

$$= 4.44 \times 50 \times 1.359 \times 10^{-3} = 0.301698\, V$$

No. of Turns:

$N_{210} = \frac{210}{0.301698} \cong 696$; $N_{22} = \frac{22}{0.301698} \cong 73$

Assuming a 90% efficiency, the currents in the different sections of the winding can be calculated, $I_{210} = \frac{A}{210 \times 0.9} = \frac{80}{210 \times 0.9} = 0.4232\, A$

From SWG wire gauge chart,

The nearest wire which can be selected above the calculated current is 23 SWG. The diameter of SWG-23 is 0.61mm, I_{max}= 0.584 A

Effective diameter considering single layer enamel coating (F class) =0.61+0.05=0.66mm

$I_{22} = \frac{A}{22 \times 0.9} = \frac{80}{22 \times 0.9} = 4.04\, A$

Similarly, The standard wire Gauge for this current is SWG-15.
Diameter=1.6256mm, I_{max}=4.15 A

Effective diameter considering single layer enamel coating (F class) =1.6256+0.05 =1.6756mm

No. of layers is therefore, $L_{22}=\frac{\frac{N_{22}}{50}}{1.05\times1.6756} \cong 3$ $L_{210}=\frac{\frac{N_{210}-N_{22}}{50}}{1.05\times0.66} \cong 9$

The Radial dimension of winding is, R = L_{22} d_{22} + L_{210} d_{210} + paper thickness • (L_{22} + L_{210} -1) + latheroid thickness • 2

$$=3 \times 1.6756 + 9 \times 0.66 + (9+3-1) \times 0.1 + 2 \times 0.3$$

$$= 12.668\ mm < 15mm$$

Open Circuit Test:

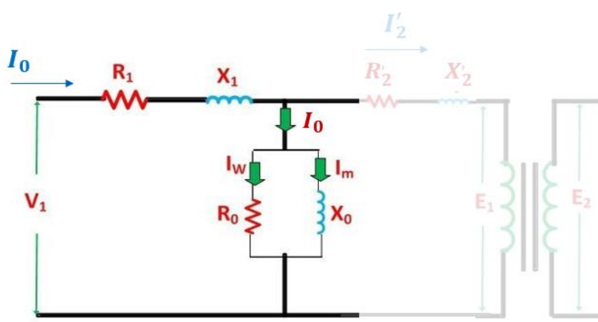


Fig 5: General equivalent circuit of transformer open circuit test

Measured value of core loss =7W

I_{LV}=0.76 A ; V(Rated)=210 V

Short Circuit Test:

Measured copper loss=4.5W;

I_{LV} =3.4 A; I_{HV} =0.42 A

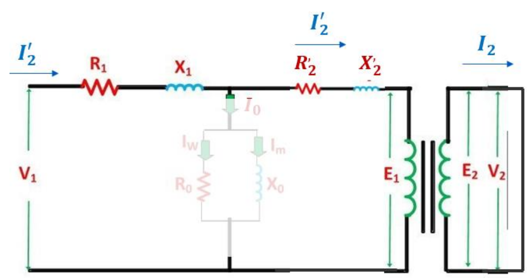


Fig 6: General equivalent circuit of transformer short circuit test

Load Test: Connecting a 200W incandescent bulb at low voltage side, we get, supplied power =16W, output power=5W,

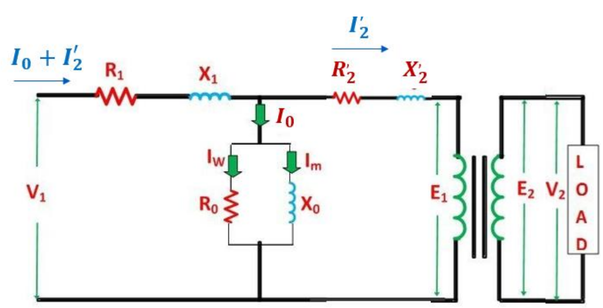


Fig 7: General equivalent circuit of transformer load test

Efficiency, $\eta = \frac{output\ power}{input\ power} = \frac{5}{16} \times 100\% = 31.25\%$ (will be around 85% to 90% if the tap was taken from a higher voltage like 100v, 150v etc.)

N:B: For autotransformer, the efficiency will increase if the tapping point is taken from a higher voltage level. This happens because, for higher voltage level current rating reduces so, copper loss also reduces increasing the overall efficiency. We would have got higher efficiency (85%-90%) if the tapping point was taken from 100-150V point.

Result & Discussion:

We have designed a 80 VA, 210V/24 V, 50 Hz, single phase auto-transformer. First of all from our given core laminations (7) we have used 64 laminations. This is due to the fact is core depth increases the cross sectional area increases. So the KVA rating will also increase.In such high KVA rating our calculated current for the taping point (lv) is significantly high & it is very difficult to adjust a standard wire gauge for that design. We just cannot simply choose a thicker wire for higher current rating because it will go out of the reel after winding. So to have a wire of within limit max current rating also satisfy the reel dimension limitation after winding, we had to reduce the KVA rating by reducing the lamination number to 64. To avoid air gap in the reel which will cause flux leakage & reduce efficiency we cut the reel to our optimized size. At first we kept the low voltage winding at the bottom & HV coil at top. But due to excessive air gap produced while winding which will eventually take the winding layers out of the reel dimensions we needed to choose plan be that is to keep the high voltage winding at the bottom layers. This method significantly reduced the air gap between layers. We couldn’t manage to give 650 turns in HV(220v) & 80 turns in LV (24v) side as required because we ran out of SWG-15 & SWG-23. We have managed to give 620 turns & 74 turns . So due to this reason our rated voltage in 210v/22v. If we still provide the 220 v supply at HV side voltage per turn will increase which will cause flux density to cross 1.2T which may permanently saturate the core. So to avoid that we had to make it a 210 v rated transformer. After out 2nd time trying it failed again because of HV internal winding short. So we had to rewind again & the 3rd time is a charm! We successfully managed to wind it this time without any issues. We measured the core & copper loss from no load & full load test where we found out the copper loss to be less than core loss which is reverse in normal types of transformers of multiple windings. This is due to during short circuit test current actually flows in a very minimal length of the full winding. So it will produce less copper loss. At last our measured efficiency is came out to be 30% which is normal for lower votage tapping point. For higher voltage tapping point the efficiency will increase a the current rating meaning the copper loss decreases with increasing voltage At 100-150v tap we would have got 85% to 90% efficiency which is expected.

Conclusion:

In conclusion we have successfully designed a compact and lightweight transformer that reduces material usage and saves space. We have ensured the transformer handles specific load requirements without saturation or overloading. We have provided customizable voltage taps for flexibility in input and output voltage levels. We also used eco-friendly materials and sustainable practices to reduce environmental impact. The purpose of this experiment has been successfully performed & we enjoyed a lot working as a team